Magnetic Properties

MO-201 Materials Science (Odd Sem 2026-27)



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Reference Textbooks

Materials Science & Engineering by V. Raghavan

Fundamentals of Materials Science & Engineering by W.D. Callister

Magnetic Materials

- **Magnetic induction or magnetic flux density B** (tesla T or Wb m^{-2}) due to a magnetic field applied in vacuum is related to the field strength H (A m^{-1}) as follows

$$B = \mu_0 H$$

where μ_0 is called the permeability of free space (vacuum). The units of permeability are

$$\frac{B}{H} = \frac{\text{Wb m}^{-2}}{\text{A m}^{-1}} \qquad (\text{Wb m}^{-2} = \text{N A}^{-1}\,\text{m}^{-1})$$

$$= \text{H m}^{-1} \qquad (\text{H} = \text{Wb A}^{-1})$$

(H m^{-1} = henry per metre)

- In SI units, the permeability of free space has a value of $4\pi \times 10^{-7}$ H m^{-1}; it was originally defined to have a value of unity in cgs units

- If the field is applied to a solid medium, the magnetic induction in the solid is given by a similar relationship

$$B = \mu H$$

where μ is the permeability of the solid material through which the magnetic lines of force pass

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- In general, μ is not equal to μ_0
- The ratio μ/μ_0 is called the **relative permeability μ_r** of the solid
- The magnetic induction in a solid is also defined by

$$B = \mu_0 (H + M)$$

where M is called the *magnetization* of the solid. The magnetization is directly related to the applied field H through the *susceptibility* of the medium χ:

$$\chi = \frac{M}{H}$$

- The orbital motion of charge carrying electrons in an atom is analogous to a current carrying coil; when a magnetic field is applied to an atom, the motion of the orbital electrons gets modified in such a way that a weak magnetic moment opposing the field is induced; **diamagnetism** is the result of this interaction
- In a diamagnetic solid, the magnetic lines of force due to an applied field are repelled
- The diamagnetic susceptibility is negative and very small in magnitude, typical values of the order of -10^{-5}

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- An important exception, however, is a superconductor, which is perfectly diamagnetic with a susceptibility value of –1; all the lines of force are repelled by superconductor, thus making them useful for the purpose of shielding out magnetic fields

- The **paramagnetic** susceptibility is small and positive, of the order of 10^{-3}; it ↓ with increasing T, as thermal energy tends to randomize the alignment

- The susceptibility is very large and positive for **ferromagnetic** materials, in the range 10^{2}–10^{5}; they strongly attract the magnetic lines of force

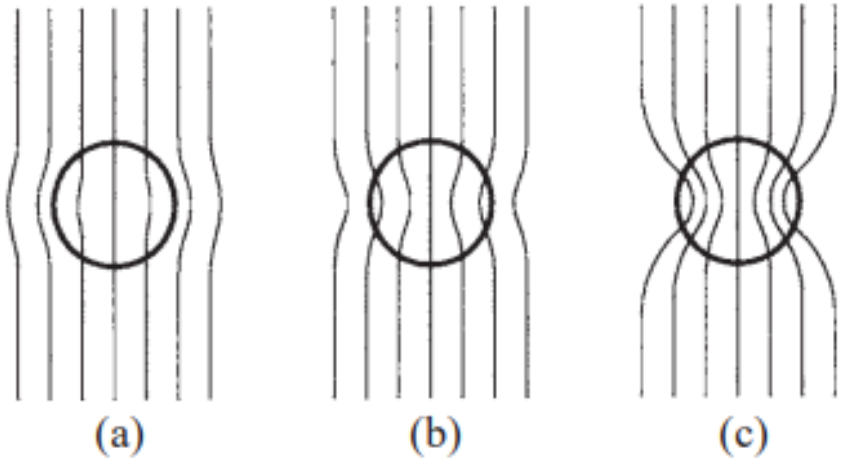


Fig. 1 (a) A diamagnetic solid repels slightly the magnetic lines of force. (b) A paramagnetic solid weakly attracts the lines of force. (c) A ferromagnetic solid attracts them very strongly

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- Some atoms and molecules possess intrinsic permanent magnetic moments; in the absence of an externally applied field, the moments of the atoms in a solid are randomly oriented with respect to one another and the solid as a whole has no net magnetic moment

- If an external field is applied, the magnetic moments tend to align themselves parallel to the applied field, so as to lower their potential energy

- Just like a suspended magnetic needle aligns itself spontaneously with the earth's magnetic field, similarly, there is a spontaneous tendency for the permanent moments of the atoms of the solid to align themselves in the direction of the field, thereby intensifying the lines of force in the field direction – this phenomenon is called **paramagnetism**

- The aligning force on the permanent moments of the atoms with ordinary magnetic fields is rather small, so that the paramagnetic effect is weak

- **Ferromagnetic** solids are those in which the permanent magnetic moments (due to electron spins) are already aligned due to bonding forces

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Magnetic Moments due to Electron Spin

- Permanent magnetic moments can arise from three sources
  - orbital magnetic moment of electrons corresponding to the quantum number m_l
  - spin magnetic moment of electrons corresponding to spin quantum number m_s
  - spin magnetic moment of the nucleus
- Of these, the spin magnetic moments of the electrons are the only ones that are important from our point of view
- An electron can be crudely approximated to a charge-carrying particle spinning about its axis and creating its own magnetic field
- The magnetic moment of an electron spin is taken as one unit, called the **Bohr magneton** μ_B (value of 9.273×10^{-24} A m^2)
- The net magnetic moment of two electrons of opposite spins is zero

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Magnetic Moments due to Electron Spin

- Atoms or molecules that have quantum states all of which have paired electrons have zero net magnetic moment

- A number of atoms and molecules, however, have unpaired electrons

- For instance, the alkali metal atoms have only one electron in the outermost s orbital; this electron is unpaired and can align itself in an applied field giving rise to paramagnetism

- The order of filling of electron orbitals in atoms is given by **Hund's rule**
  - All orbitals will be singly occupied before any is doubly occupied
  - All of the unpaired electrons must have the same spin

- An atom with 3 electrons in a p orbital will have all 3 spins aligned, giving rise to a net magnetic moment of 3 μ_B; on the other hand, an atom with 4 electrons in the p orbital will have a net moment of only 2 μ_B, as the spin of the fourth electron is opposite to that of the first three; Similarly, an atom with 5 electrons in a d orbital has a net magnetic moment of five units; but an atom with 9 electrons in the d orbital has a net moment of only one; 7 electrons in f orbital will give a net magnetic moment of 7 units

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Magnetic Moments due to Electron Spin

- This simple deduction of the net magnetic moment of an atom has to be modified in the solid state, where the electron energy levels of the outermost orbitals interact and form energy bands

- In the transition metals, the energy bands due to the 4s orbital and the 3d orbital may overlap to some extent; the 4s electrons spend some time in the 3d band; if the 3d electrons are five or more in number, the spins of the overlapping 4s electrons must be opposite to that of the unpaired 3d electrons – such an overlap, therefore, reduces the net magnetic moment

- The Fe atom with an outer electronic configuration of $3d^6 4s^2$ has a moment of 4 units as a free atom, but has a moment of only 2.2 as part of the crystal

- Similarly, Co with a configuration of $3d^7 4s^2$ has a moment of 1.7 and Ni with a configuration of $3d^8 4s^2$ has a moment of only 0.6 unit in the crystal

- In the case of Gd, the net magnetic moment of seven in the free atom remains unchanged in the solid state, as there is no overlap of the 4f orbitals with the other energy bands

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Magnetic Moments due to Electron Spin

- The magnetization of a solid is the sum of the magnetic moments in unit volume of the solid

- Magnetization = magnetic moments/volume
  $= A\ m^2/m^3$
  $= A\ m^{-1}$

Example The saturation magnetization of BCC iron is 1750 kA m^{-1}. Calculate the net magnetic moment per iron atom in the crystal.

Solution

The lattice parameter of BCC iron	$= 2.87$ Å
Volume of the unit cell	$= 2.87^3 \times 10^{-30}\ m^3$
Number of atoms in the unit cell	$= 2$
Net magnetic moment per atom	$= 1750 \times 1000 \times 2.87^3 \times 10^{-30} \times 1/2$
	$= 2.068 \times 10^{-23}\ A\ m^2$
In units of μ_B, the moment	$= 2.068 \times 10^{-23}/(9.273 \times 10^{-24})$
	$= 2.2$

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Ferromagnetism and Related Phenomena

- Hund's rule is a manifestation of the spin dependent electrostatic energy of the electrons in an orbital
- When the electrons have the same momentum and the same spin, a certain distance must separate them from one another, in order to be consistent with the Pauli exclusion principle
- This physical separation tends to increase their kinetic energy, but reduces the electrostatic repulsive energy between them
- Whether the electrons will align their spins or not will depend on the sign of the net change in energy
- The stable, energetically favoured order of filling of electronic orbitals in atoms is given by Hund's rule

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Ferromagnetism and Related Phenomena

- In the solid state, the outer electronic orbitals of neighbouring atoms overlap and produce energy bands
- Consider the overlapping and the formation of energy bands in the first transition metals
- At the left end of the series, the radius of the 3d orbitals in the atoms is large enough to result in good overlap between neighbours in the crystalline state
- The 3d band then contains all paired up electrons and there is no net magnetic moment in the crystal
- We can say that the atoms are **antiferromagnetically coupled**, as the magnetic moments of a pair of atoms exactly cancel out

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Ferromagnetism and Related Phenomena

- As we go to the right in the transition series, the 3d orbitals shrink due to greater attraction from the increasing charge on the nucleus and consequently the overlap of the 3d orbitals decreases

- The elements Fe, Co and Ni are in special situation; the unpaired electrons in the 3d orbitals of neighbouring atoms align their spins in a parallel fashion and thereby lower their spin dependent electrostatic energy

- This lowering is partly offset by the rise in the Fermi level and the consequent increase in the average kinetic energy of the electrons

- The increase in the Fermi level is directly attributable to the greater physical separation between electrons

- The net gain in energy, $E_{unmagnetized} - E_{magnetized}$, called the **exchange interaction energy** is a sensitive function of the ratio of the atomic diameter to the 3d orbital diameter

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Ferromagnetism and Related Phenomena

Element	Ti	Cr	Mn	Fe	Co	Ni
$d_{\text{atomic}}/d_{3d\ \text{orbital}}$	1.12	1.18	1.47	1.63	1.82	1.98

- It turns out that, only when this ratio lies between 1.5 and 2.0, the exchange energy is positive and parallel spins are energetically favoured

- Among the common metals, only Fe, Co and Ni have positive exchange energy and are in the spontaneously magnetized state; these metals or their ions form the basis of the majority of magnetic materials

- No overlap of the d orbitals occurs in the higher transition series, as the d electrons are strongly attracted to their respective nuclei, which have larger positive charges

- The only other elemental crystals where the exchange energy is appreciable belong to the first rare earth series such as gadolinium, terbium and dysprosium – here, the 4f electrons align themselves in a parallel fashion

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Ferromagnetism and Related Phenomena

- Thermal energy tends to randomize the aligned spins, so that all ferromagnetic materials become paramagnetic at a sufficiently high temperature
- The transition temperature at which all the spin alignment is lost is called the **Curie temperature**
- Just below the Curie temperature, the susceptibility can be 10^2 to 10^3, increasing with decrease in temperature, as the alignment becomes more complete
- The Curie temperature is a function of the magnitude of the exchange energy
- Co that has the highest exchange energy has the highest Curie temperature, 1400 K
- Gd with a small exchange energy has a Curie temperature below room temperature

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Ferromagnetism and Related Phenomena

- The interatomic distance can be changed within limits by alloying
- For example, Mn in the elemental form is not ferromagnetic
- On alloying, the exchange energy can become positive and the spins on neighbouring atoms can become aligned if the Mn–Mn distance is increased by the right amount, so that the ratio of the atomic diameter to the 3d orbital diameter falls in the range 1.5–2.0
- This happens in *Heusler alloys* Cu_2MnSn and Cu_2MnAl, which are ferromagnetic

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Ferromagnetism and Related Phenomena

- In some compounds, the constituent atoms may be antiferromagnetically coupled, but can have different magnetic moments

- This would give rise to a net magnetic moment in each coupling and the sum of the moments of all the couplings can result in magnetization that is comparable in order of magnitude to ferromagnetism

- This phenomenon is called **ferrimagnetism** and is compared with ferromagnetism and antiferromagnetism in Fig. 2

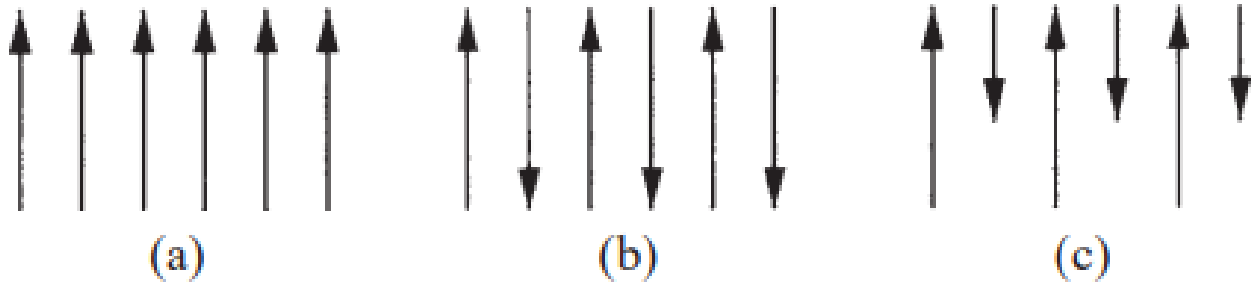


Fig. 2 (a) Ferromagnetic, (b) antiferromagnetic, and (c) ferrimagnetic coupling of electron spins in atoms denoted by arrows; the length of an arrow is a measure of the magnetic moment of the atom

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Ferromagnetism and Related Phenomena

- Ferrites of the general formula $Me^{2+}Fe_2^{3+}O_4^{2-}$ can be ferrimagnetic
- The divalent metal cation (Me^{2+}) in the formula is usually Fe^{2+}, Ni^{2+}, Zn^{2+}, Mg^{2+}, Co^{2+}, Ba^{2+}, Mn^{2+} or some combination of these ions
- These compounds crystallize mostly as the *inverse spinel structure* – different cation distribution in that all of the A cations and half of the B cations occupy octahedral sites, while the other half of the B cations occupy tetrahedral sites (AB_2O_4)
- As an example of the incomplete cancellation of the magnetic moments, consider the crystal of magnetite, Fe_3O_4 – the ferric ion, with two 4s electrons and one 3d electron removed from the neutral iron atom, has a magnetic moment of 5 units and the ferrous ion has 4 units
- All the ferric ions in magnetite are antiferromagnetically coupled, so that the net magnetic moment of the compound arises from the ferrous ions only
- The magnetization M calculated on the basis of 4 net magnetic moments per Fe_3O_4 is in good agreement with the experimental value for magnetite (ferrimagnetic)

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Ferromagnetism and Related Phenomena

Example In nickel ferrite, $NiFe_2O_4$, the ferric ions are antiferromagnetically coupled. The magnetization is due to the nickel ions. When zinc is added to nickel ferrite, the magnetization of the crystal increases, even though the zinc ions are not ferromagnetic. Explain how this could happen.

Solution the inverse spinel structure of nickel ferrite has all the Ni^{2+} and half of the Fe^{3+} ions in the octahedral sites. The other half of the Fe^{3+} ions are in the tetrahedral sites. The ferric ions in the two sites are antiferromagnetically coupled. When zinc substitutes for the ferric ions in the tetrahedral sites, the antiferromagnetic coupling will not exactly cancel, as the number of ferric ions in the octahedral sites will be more. This would increase the magnetization, as there is an extra contribution from the uncoupled ferric ions of the octahedral sites.

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The Domain Structure

- Iron has a high Curie temperature of 1041 K (768°C) and, consequently, almost all the spins remain aligned at room temperature

- Yet, at room temperature, an ordinary piece of iron is not magnetic in the absence of an applied field; to explain this discrepancy, Weiss conceived the idea of **magnetic domains**

- The unmagnetized iron crystal consists of a number of domains and within each domain, the spins of all the atoms are aligned; however, the spins of adjacent domains are not parallel; due to the random orientation of the domains with respect to one another, the magnetic moments cancel out and the crystal as a whole is not magnetic

- In the presence of an external field, the domains tend to align themselves with the field, resulting in a large net magnetization of the solid

- The process of magnetization of the iron crystal is not the aligning of the electron spins of the different atoms, but it is the alignment of the various domains, each of which is already ferromagnetic

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The Domain Structure

- The magnetostatic energy of a ferromagnetic solid can be reduced if a number of domains are arranged such that no poles exist at the surface and no lines of force go out of the material, as illustrated in Fig. 3

- Hence, a single domain tends to break up into several domains, even if this means an increase in the domain boundary energy; the domain size does not ↓ indefinitely, as at some stage the increasing domain boundary energy would oppose this

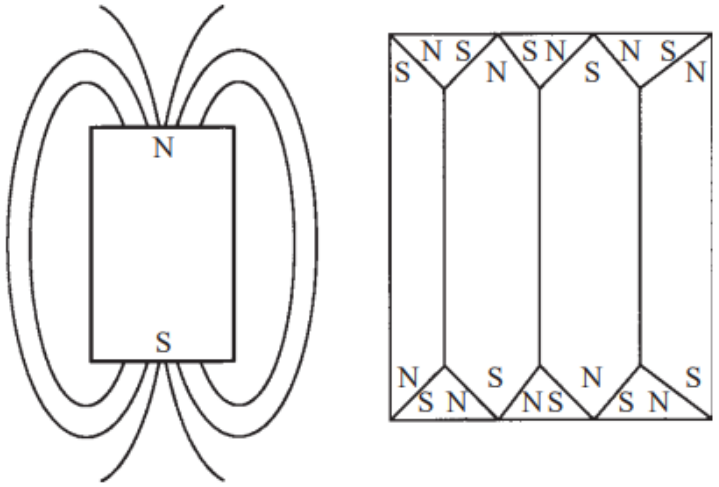


Fig. 3 (a) With one domain, the magnetic lines of force go out of the material (b) The magnetic lines do not go out if a number of domains are present

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The Hysteresis Loop

- The B-H curve for a typical ferromagnetic material is shown in Fig. 4
- As the applied field H is increased, the magnetic induction B increases slowly at first and then more rapidly
- The rate of magnetic induction slows down again, eventually attaining a *saturation value* B_s
- With further increase in the magnetic field, there is no increase in the induction
- If the field is reversed, the induction decreases slowly at first and reaches a residual value B_r at zero field

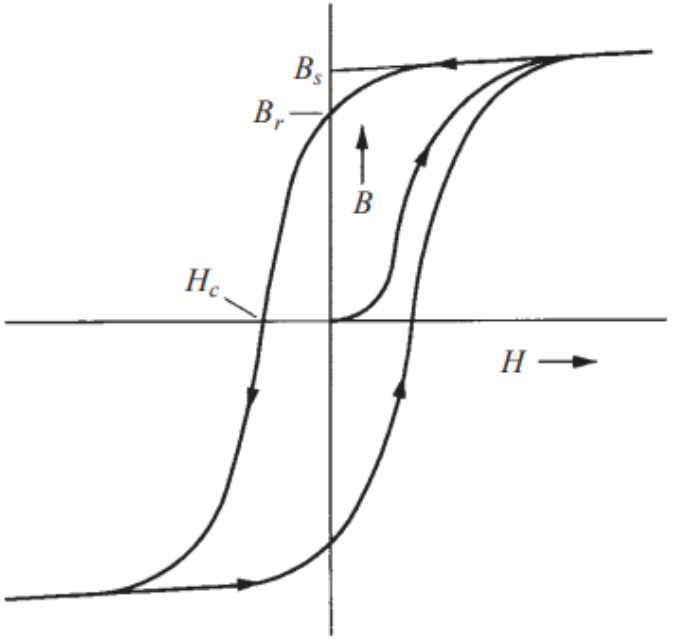


Fig. 4 Magnetic induction B as a function of the applied field H for a ferromagnetic material, tracing a hysteresis loop

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The Hysteresis Loop

- B_r represents the amount of *residual induction* left in the specimen after the removal of the field
- If the application of the field is continued in the opposite direction, the domains tend to reverse their alignment, so that the remaining induction is lost at a certain value of the reverse field called the **coercive field** H_c
- The process of reversal of domains continues to give a net magnetization in the opposite direction
- After saturation occurs in this direction, restoring the original field direction completes the **hysteresis loop**

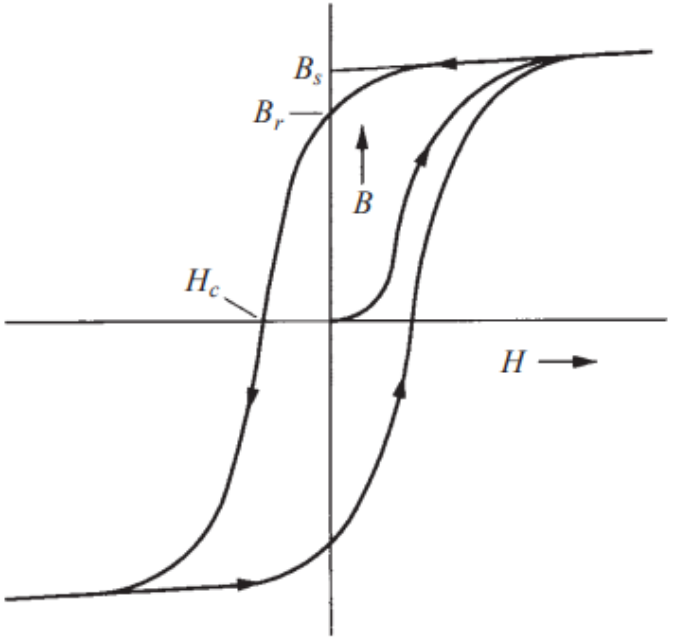


Fig. 4 Magnetic induction B as a function of the applied field H for a ferromagnetic material, tracing a hysteresis loop

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The Hysteresis Loop

- There are two possible ways to align a random domain structure by applying an external field
- One is to rotate a domain in the direction of the field and the other is to allow the growth of the more favourably oriented domains at the expense of the less favourably oriented ones
- If we compare the domain structure with the grain structure of a polycrystalline material, the boundaries separating the domains called domain walls are the analogue of grain boundaries
- The domain boundary energy is about 0.002 J m^{-2}; the domain walls, however, are some two orders of magnitude thicker than the grain boundaries, because there is a gradual transition from one domain orientation to the next across the wall
- Also, the domain boundaries can exist within the grain

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The Hysteresis Loop

- Analogous to grain growth, the domain walls can move such that the more favourably oriented domains grow at the expense of others
- In the earlier stages of magnetization below the saturation region of the hysteresis curve, domain growth is dominant
- The growth is more or less complete as the saturation region is approached
- Thereafter, the most favourably oriented, fully grown domain tends to rotate so as to be in complete alignment with the field direction
- The energy required to rotate an entire domain is more than that required to move the domain walls during growth; consequently, the slope of the B-H curve decreases on approaching saturation
- Each time the hysteresis loop is traversed, energy equal to the area of the loop is dissipated as heat; the power loss due to hysteresis in a transformer core is dependent on the number of times the full loop is traversed per second

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Soft and Hard Magnetic Materials

- **Soft magnetic materials** are used in applications requiring frequent reversals of the direction of magnetization

- In soft magnetic materials, the hysteresis losses must be kept down to a minimum

- When the induction is large for a small applied field, the loop area is small and the hysteresis loss is reduced; the key factor in the design of a soft magnet is then to have easily moving domain walls

- **Hard magnetic materials** are used to produce permanent magnets; hysteresis losses are of no significance here, as no repeated reversals of the magnetization is involved in a permanent magnet; they must have a high residual induction B_r and a large coercive force H_c

- The area of the hysteresis loop between B_r and H_c represents the energy required to demagnetize a permanent magnet; the maximum value of this area B_rH_c, called the energy product, must be as large as possible for permanent magnets

- High C steels and other low-alloy tungsten and chromium steels are used in the martensitic condition as permanent magnets